SRI CHANDRASEKHARENDRA SARASWATHI VISWA MAHAVIDYALAYA

(UNIVERSITY ESTABLISHED UNDER SECTION 3 OF UGC ACT 1956)

ENATHUR, KANCHIPURAM – 631 561

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

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Class: II B.E. (CSE)

Course Code:

Course Name: Data Science Toolkit Lab

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BONAFIDE CERTIFICATE

This is to certify that this is the bonafide record of work done by **Mr/Ms. RAMACHANDRUNI VALLI HARIKA** with **Reg. No :11249A311** of II-B.E.(CSE) during the academic year 2025 – 2026.

Station:**ENATHUR**

Date:

Staff-in-charge Head of the Department

Submitted for the Practical examination held on:

Examiner-1 Examiner-2

INDEX

SL.NO	Program Name	Page no	Date	Sign
1.1	**NUMPY**	**3-8**	**22-1-26**	
1.2	**PANDAS**	**9-11**	**29-1-26**	
1.3	**MATPLOTLIB**	**12-15**	**5-2-26**	
1.4	**SCIKIT-LEARN**	**16-19**	**12-2-26**	
2	**Perform Data Exploration and preprocessing in Python**	**20-24**	**12-2-26**	
3	**Implement Regularised Linear Regression**	**25-26**	**19-2-26**	
4	**Implement Naive Bayes Classifier**	**27-29**	**5-3-26**	
5	**Implement Regularised Logistic Regression**	**30-33**	**5-3-26**	
6	**Build models using different Ensembling Techniques**	**34-36**	**12-3-26**	
7	**Build models using decision Trees**	**37-38**	**19-3-26**	
8	**Support Vector Machine (SVM) with different kernels**	**39-41**	**3-4-26**	
9	**Implement K-Nearest Neighbors (KNN) Classification**	**42-44**	**3-4-26**	
10	**K-means clustering with PCA and ELBOW method**	**45-48**	**10-4-26**	

EX-01

1.NUMPY(LIBRARY)

```
import numpy as np

# Create a list and convert to NumPy array
list1 = [1, 2, 3]
vector1 = np.array(list1)
print("Vector 1:", vector1)

# Create a range array
vector = np.arange(1, 6)
print("Arange vector:", vector)

# Create zeros array
vector = np.zeros(5)
print("Zeros:", vector)

# Dot product example
list1 = [5, 1, 1]
list2 = [3, 2, 1]

vector1 = np.array(list1)
vector2 = np.array(list2)

print("Vector 1:", vector1)
print("Vector 2:", vector2)

product = vector1.dot(vector2)
print("Dot product:", product)

# More NumPy examples
a = np.array([1, 2, 3, 4, 5])
b = np.arange(1, 10, 2)
c = np.linspace(0, 1, 5)
d = np.zeros((2, 3))
e = np.ones((2, 2))
f = np.eye(3)
g = np.random.randint(1, 100, size=(3, 3))

print("a:", a)
print("b:", b)
print("c:", c)
print("d:\n", d)
print("e:\n", e)
print("f (identity matrix):\n", f)
print("g (random):\n", g)
```

2)Array properties

```
print("Shape:", g.shape)
print("Dimensions:", g.ndim)
print("Size:", g.size)
print("Data type:", g.dtype)
```

3) Reshape & Transpose

```
reshaped = g.reshape(1, 12)
print("Reshaped array:\n", reshaped)
print("Transpose:\n", g.T)
print("Flattened:", g.flatten())
```

4) Mathematical Operations

```
print("Add:", np.add(a, 2))
print("Multiply:", np.multiply(a, 3))
print("Square root:", np.sqrt(a))
print("Power:", np.power(a, 2))
print("Absolute:", np.abs([-2, 3]))
```

5) Statistical Functions

```
print("Mean:", np.mean(a))
print("Median:", np.median(a))
print("Standard Deviation:", np.std(a))
print("Variance:", np.var(a))
print("Minimum:", np.min(a))
print("Maximum:", np.max(a))
print("Sum:", np.sum(a))
print("Percentile (50):", np.percentile(a, 50))
```

6)Axis-based operations

```
print("Column-wise sum:\n", np.sum(g, axis=0))
print("Row-wise mean:\n", np.mean(g, axis=1))
```

7) Indexing & Slicing

```
print("First element:", a[0])
print("Slice:", a[1:4])
print("2D Slice:\n", g[0:2, 1:3])
```

8)Boolean & Conditional Operations

```
print("Where > 50:\n", np.where(g > 50))
print("Any > 90:", np.any(g > 90))
print("All > 0:", np.all(g > 0))
```

9)Searching + Sorting

```
print("Sorted a:", np.sort(a))
print("Index of max:", np.argmax(a))
print("Index of min:", np.argmin(a))
print("Unique elements:", np.unique([1, 2, 2, 3, 4, 4]))
```

10)Linear Algebra

```
x = np.array([[1, 2],[3, 4]])
y = np.array([[5, 6],[7, 8]])

print("Dot Product:\n", np.dot(x, y))
print("Matrix Multiplication:\n", np.matmul(x, y))
print("Determinant:", np.linalg.det(x))
print("Inverse:\n", np.linalg.inv(x))
```

11)Random Functi

```
import numpy as np
np.random.seed(1)
a = np.array([1, 2, 3, 4, 5])
rand_sample = np.random.choice(a, size=3)
print("Random choice:", rand_sample)
```

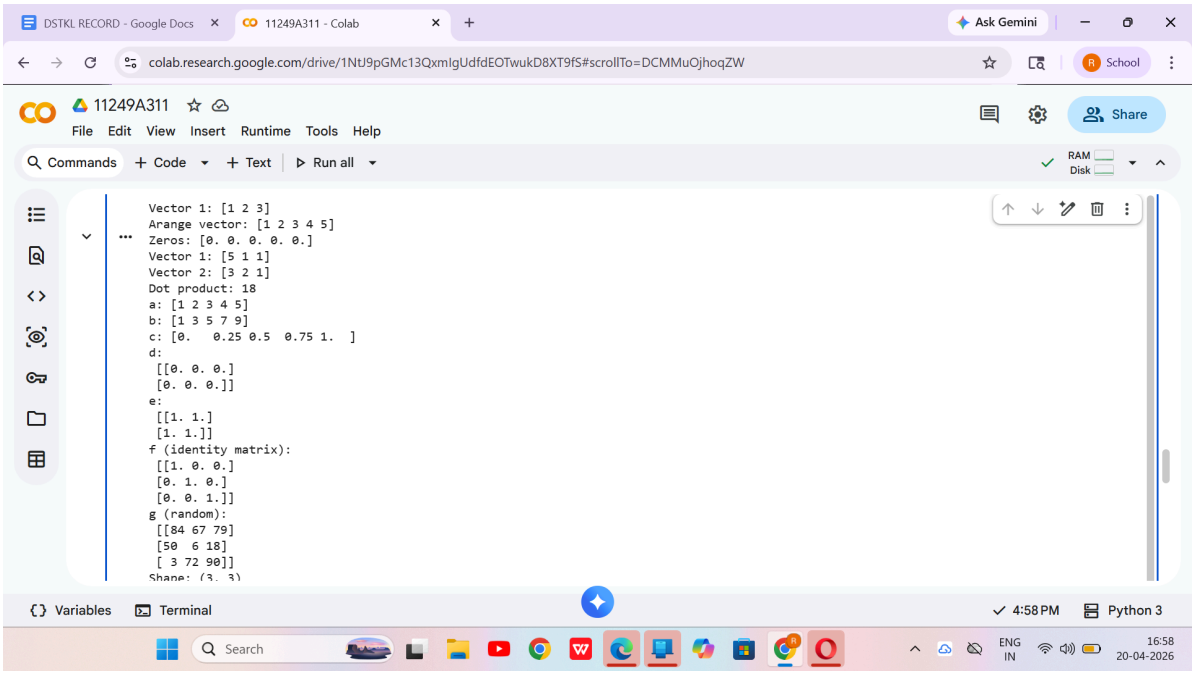
12)

```
# Create arrays
h1 = np.array([1, 2, 3])
h2 = np.array([4, 5, 6])

# Concatenate and Stack
print("Concatenate:", np.concatenate((h1, h2)))
print("Vertical Stack:\n", np.vstack((h1, h2)))
print("Horizontal Stack:", np.hstack((h1, h2)))
import numpy as np
```

13)

```
# Save array to file
np.save("sample-array.npy", a)
# Load array from file
loaded = np.load("sample-array.npy")
print("Loaded array:", loaded)
```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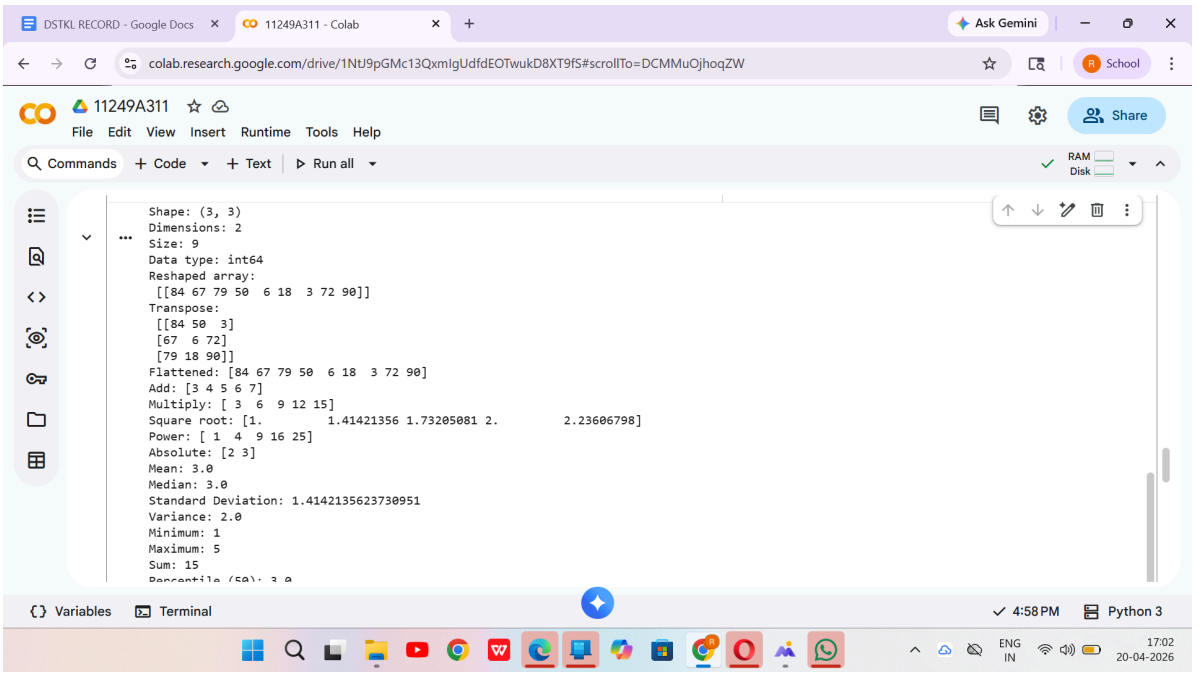
RAM Disk
Vector 1: [1 2 3]
Arange vector: [1 2 3 4 5]
Zeros: [0. 0. 0. 0. 0.]
Vector 1: [5 1 1]
Vector 2: [3 2 1]
Dot product: 18
a: [1 2 3 4 5]
b: [1 3 5 7 9]
c: [0. 0.25 0.5 0.75 1.]
d:
[[0. 0. 0.]
[0. 0. 0.]]
e:
[[1. 1.]
[1. 1.]]
f (identity matrix):
[[1. 0. 0.]
[0. 1. 0.]
[0. 0. 1.]]
g (random):
[[84 67 79]
[50 6 18]
[3 72 90]]
Shape: (3, 3)
Variables
Terminal
4:58 PM
Python 3
Search
ENG IN
16:58
20-04-2026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School
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Commands + Code + Text Run all
RAM Disk
Shape: (3, 3)
Dimensions: 2
Size: 9
Data type: int64
Reshaped array:
[[84 67 79 50 6 18 3 72 90]]
Transpose:
[[84 50 3]
[67 6 72]
[79 18 90]]
Flattened: [84 67 79 50 6 18 3 72 90]
Add: [3 4 5 6 7]
Multiply: [3 6 9 12 15]
Square root: [1. 1.41421356 1.73205081 2. 2.23606798]
Power: [1 4 9 16 25]
Absolute: [2 3]
Mean: 3.0
Median: 3.0
Standard Deviation: 1.4142135623730951
Variance: 2.0
Minimum: 1
Maximum: 5
Sum: 15
Variables
Terminal
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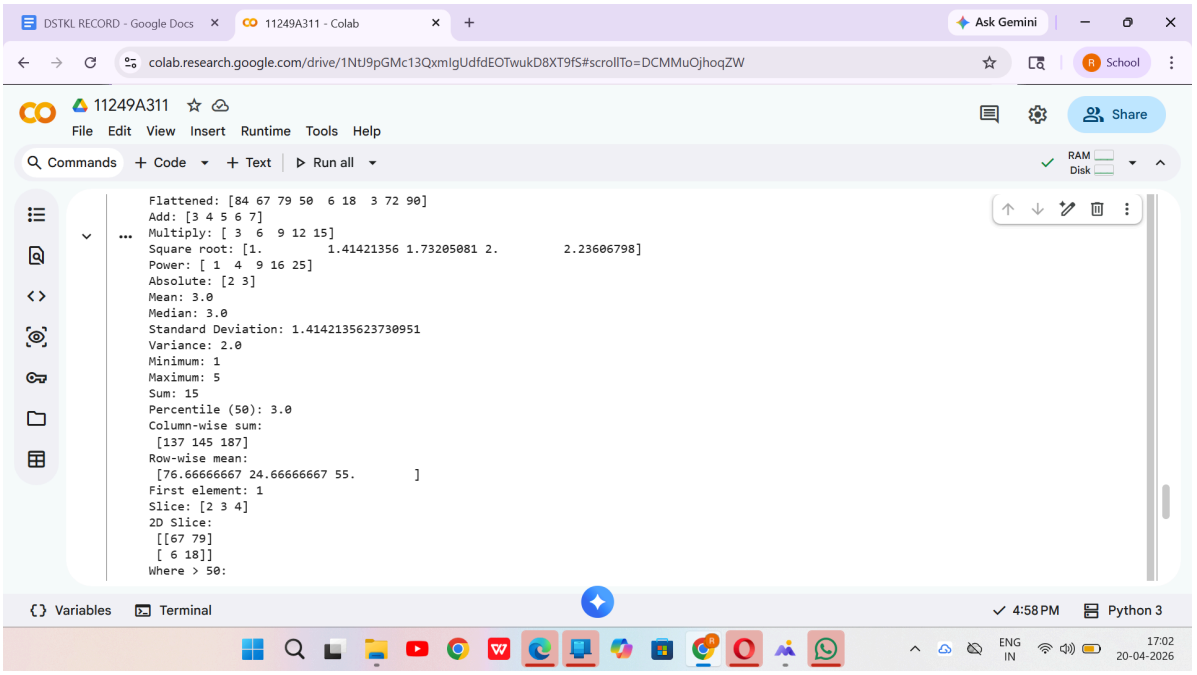
20-04-2026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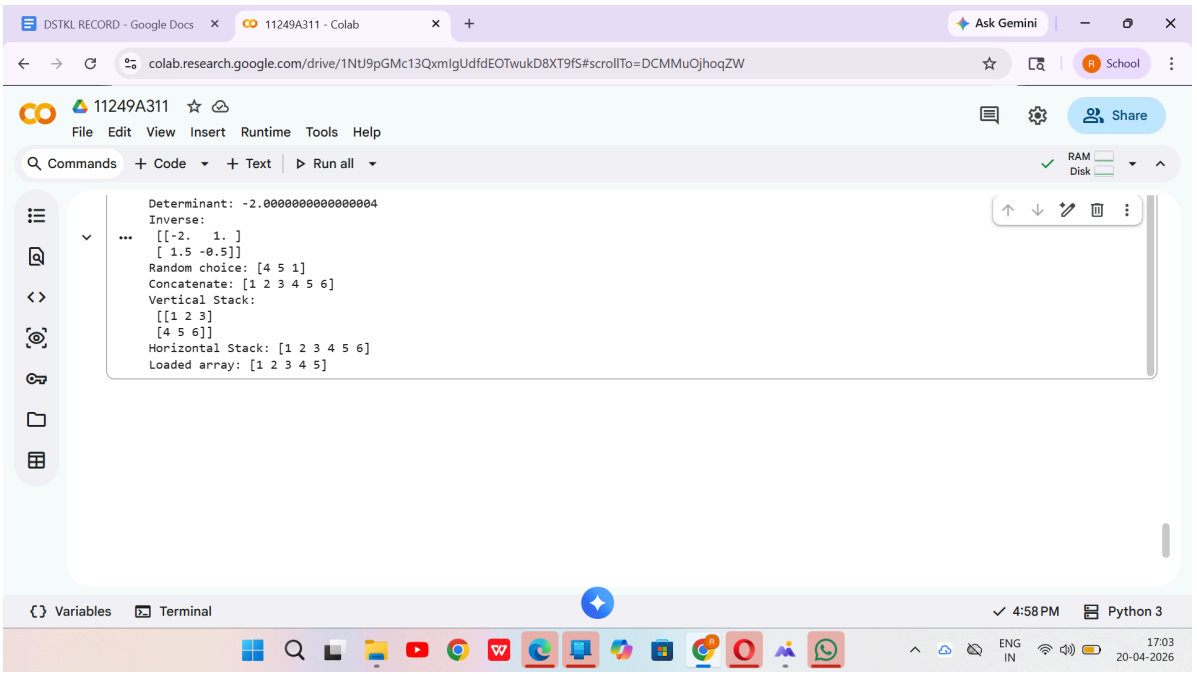
RAM Disk
Flattened: [84 67 79 50 6 18 3 72 90]
Add: [3 4 5 6 7]
Multiply: [3 6 9 12 15]
Square root: [1. 1.41421356 1.73205081 2. 2.23606798]
Power: [1 4 9 16 25]
Absolute: [2 3]
Mean: 3.0
Median: 3.0
Standard Deviation: 1.4142135623730951
Variance: 2.0
Minimum: 1
Maximum: 5
Sum: 15
Percentile (50): 3.0
Column-wise sum:
[137 145 187]
Row-wise mean:
[76.66666667 24.66666667 55.]
First element: 1
Slice: [2 3 4]
2D Slice:
[[67 79]
[6 18]]
Where > 50:
Variables Terminal
4:58 PM Python 3
ENG IN
17:02
20-04-2026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Commands + Code + Text Run all
RAM Disk
Determinant: -2.0000000000000004
Inverse:
[[-2. 1.]
[1.5 -0.5]]
Random choice: [4 5 1]
Concatenate: [1 2 3 4 5 6]
Vertical Stack:
[[1 2 3]
[4 5 6]]
Horizontal Stack: [1 2 3 4 5 6]
Loaded array: [1 2 3 4 5]
Variables Terminal
4:58 PM Python 3
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17:03
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PANDAS

Aim:

To use Pandas for data manipulation and analysis, including tasks such as data cleaning, transformation, and handling structured datasets efficiently.

```
import pandas as pd
```

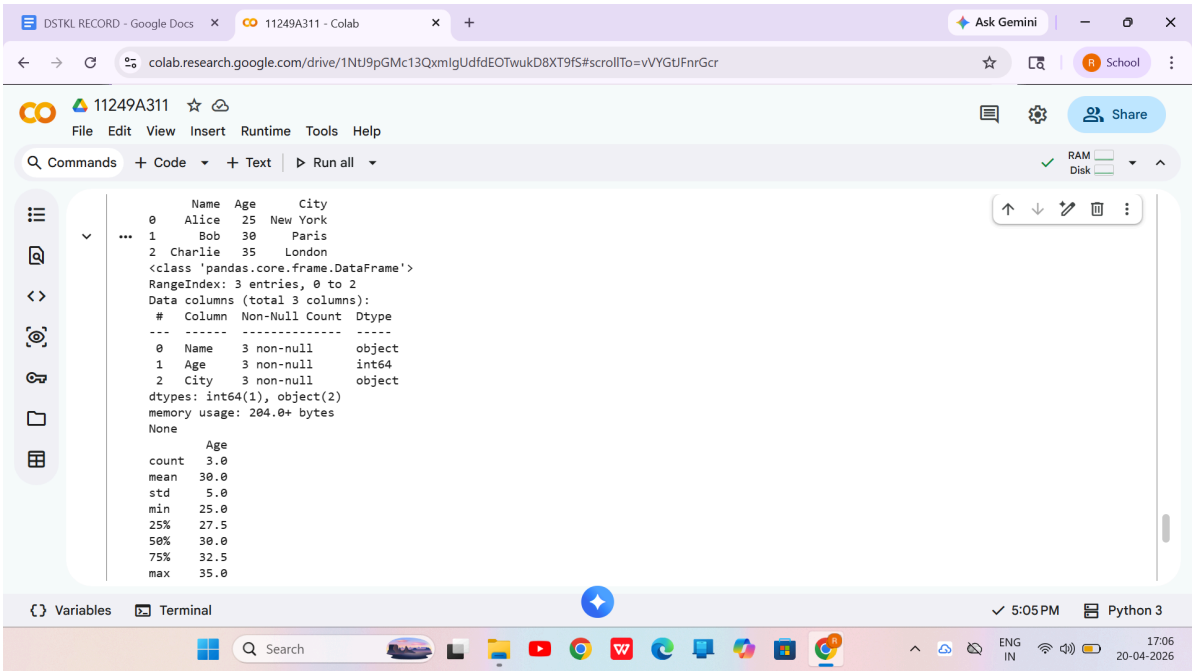
A) Creating DataFrame

```
data = {
"Name": ["Alice", "Bob", "Charlie"],
"Age": [25, 30, 35],
"City": ["New York", "Paris", "London"]
}

df = pd.DataFrame(data)


print(df.head())
print(df.info())
print(df.describe())
```

OUTPUT:



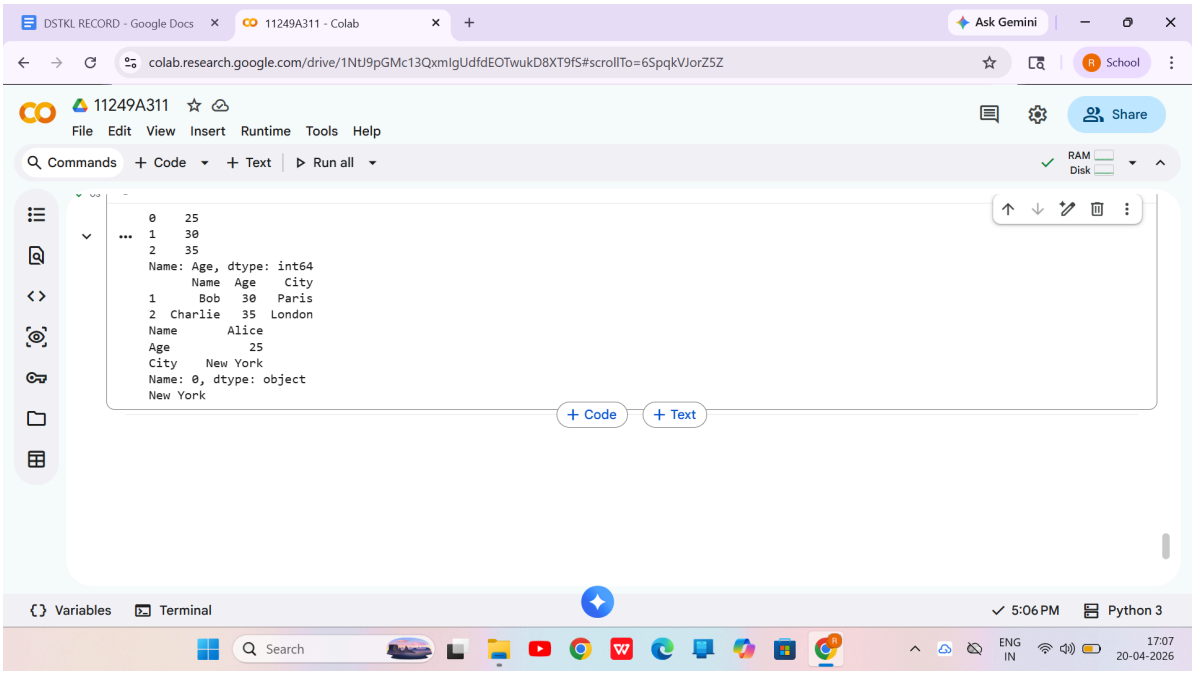
B) Selecting + Filtering

```
ages = df["Age"]
above_25 = df[df["Age"] > 25]
row_0 = df.iloc[0]

# Specific value (row 0, City column)
specific_val = df.loc[0, "City"]

print(ages)
print(above_25)
print(row_0)
print(specific_val)
```

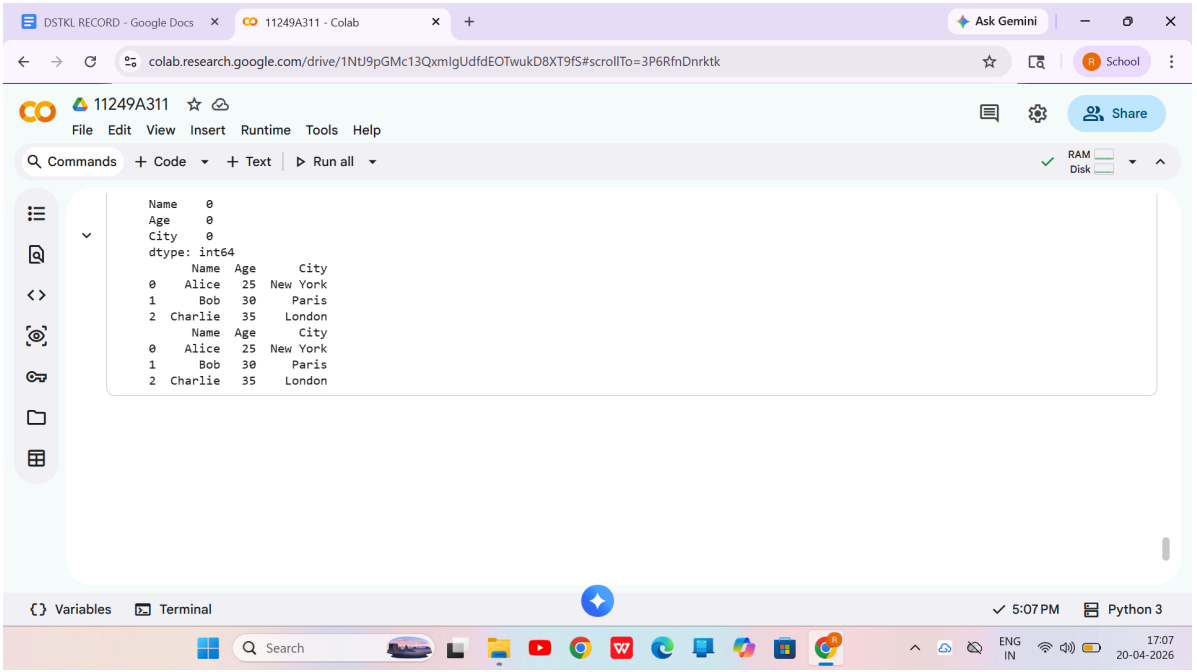
OUTPUT:



C) Data Cleaning (Handling Missing Values)

```
print(df.isnull().sum())
df_clean = df.dropna()
df_filled = df.fillna(0)
# Fill missing Age values with mean
df["Age"] = df["Age"].fillna(df["Age"].mean())
print(df_clean)
print(df_filled)
```

OUTPUT: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Commands + Code + Text Run all
RAM
Disk
Name 0
Age 0
City 0
dtype: int64
Name Age City
0 Alice 25 New York
1 Bob 30 Paris
2 Charlie 35 London
Name Age City
0 Alice 25 New York
1 Bob 30 Paris
2 Charlie 35 London
Variables
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20-04-2026

MATPLOTLIB(plotting library)

Aim:

To use Matplotlib for creating visualizations such as line plots, bar charts, and scatter plots to analyze and interpret data effectively.

Program:

```
import pandas as pd

import matplotlib.pyplot as plt

import numpy as np

data = {

"StudyHours": [1, 2, 3, 4, 5, 6, 7, 8],

"ExamScore": [35, 40, 50, 55, 65, 70, 78, 85]

}

df = pd.DataFrame(data)
```

1) Scatter Plot

```
plt.scatter(df["StudyHours"], df["ExamScore"])

plt.xlabel("Study Hours")

plt.ylabel("Exam Score")

plt.title("Study Hours vs Exam Score (Scatter Plot)")
```

```
plt.show()
```

2) Line Plot

```
plt.plot(df["StudyHours"], df["ExamScore"], marker='o')

plt.xlabel("Study Hours")

plt.ylabel("Exam Score")

plt.title("Exam Score Progress (Line Plot)")

plt.show()
```

3) Histogram

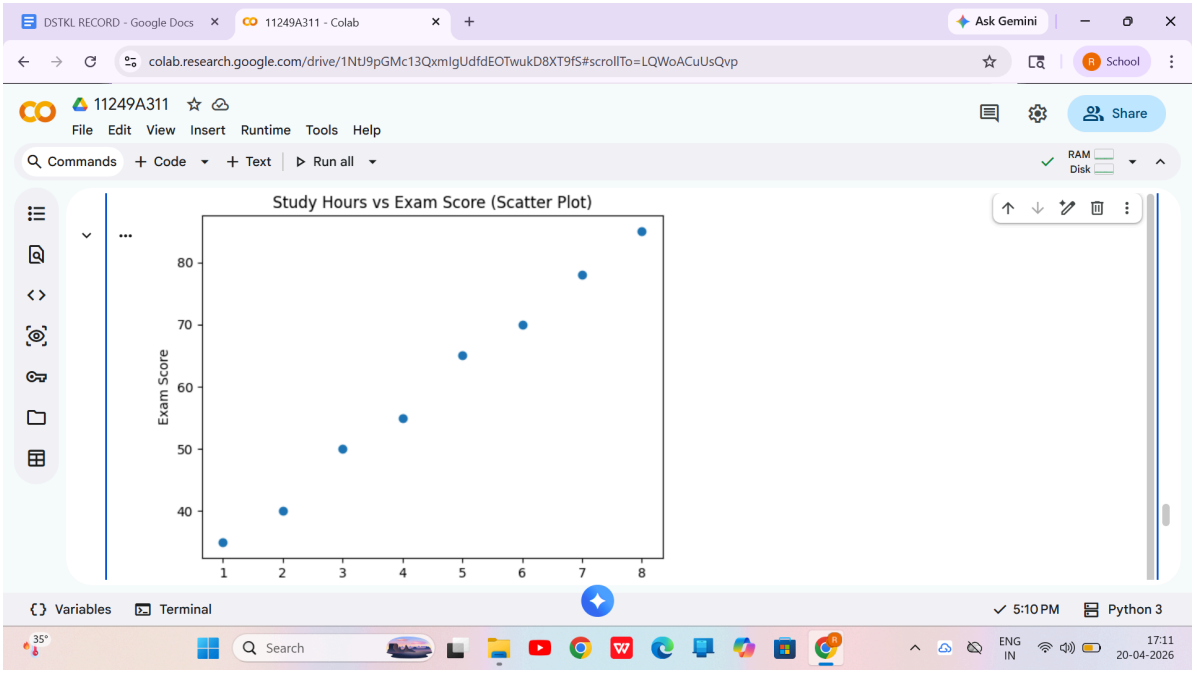
```
plt.hist(df["ExamScore"], bins=5)

plt.xlabel("Score Range")

plt.ylabel("Number of Students")

plt.title("Exam Score Distribution (Histogram)")

plt.show()
```

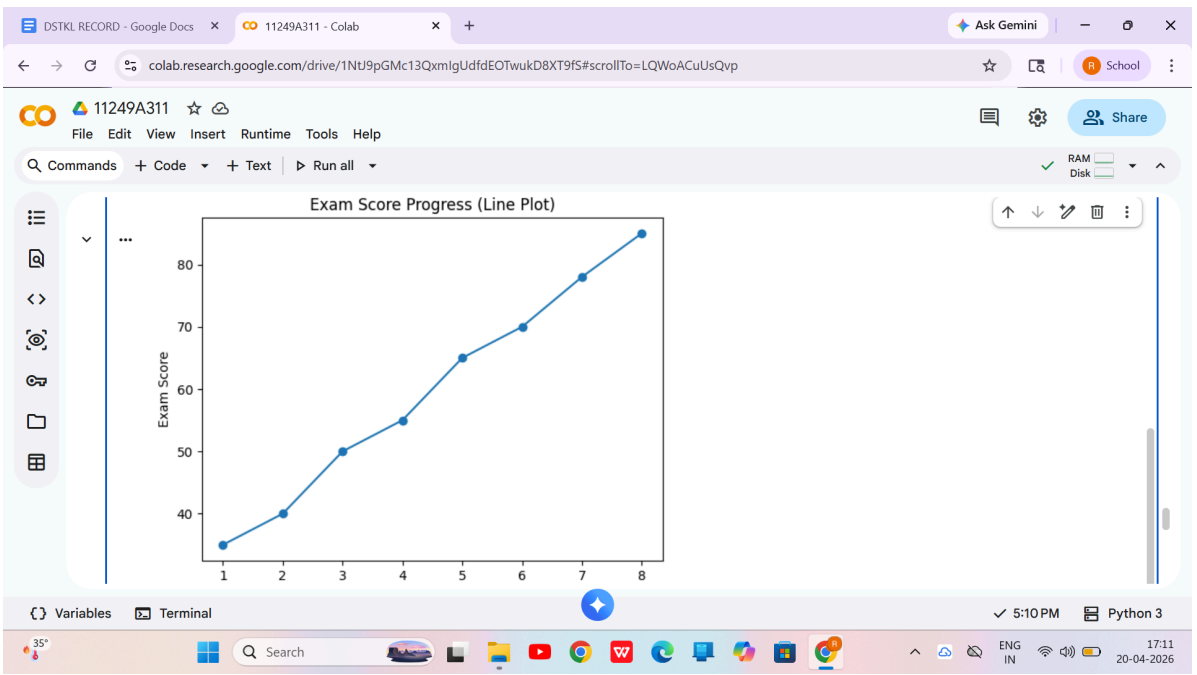
4) Bar Chart

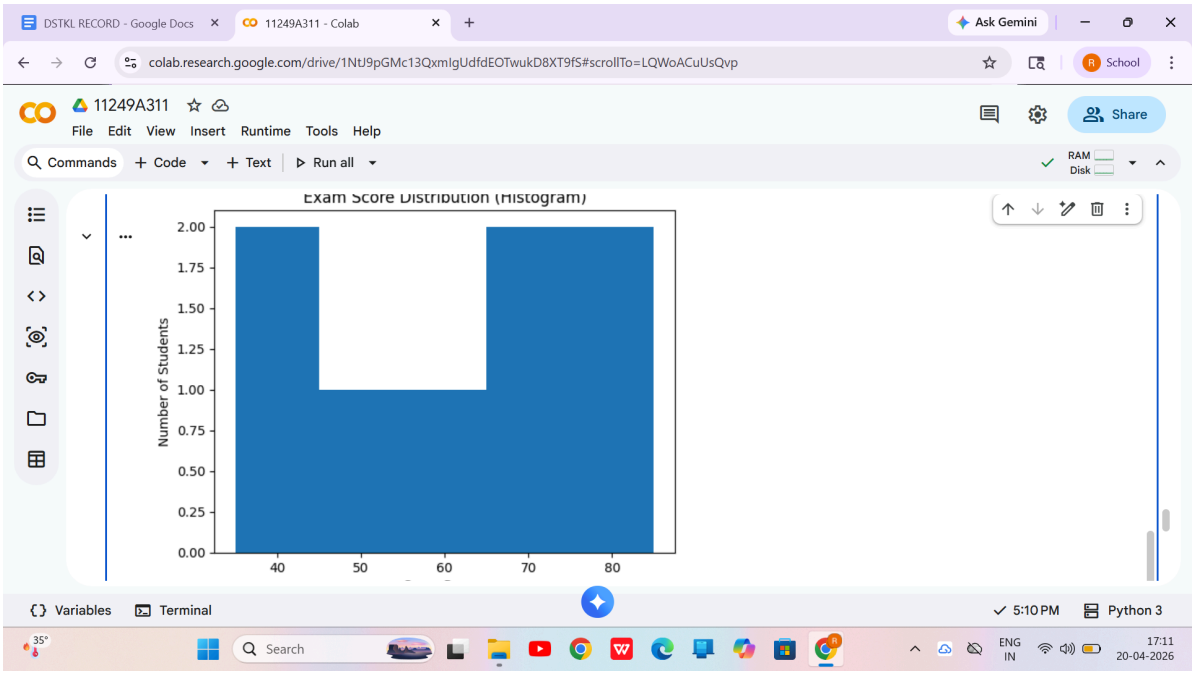
```
plt.bar(df["StudyHours"], df["ExamScore"])

plt.xlabel("Study Hours")

plt.ylabel("Exam Score")

plt.title("Study Hours vs Exam Score (Bar Chart)")
```

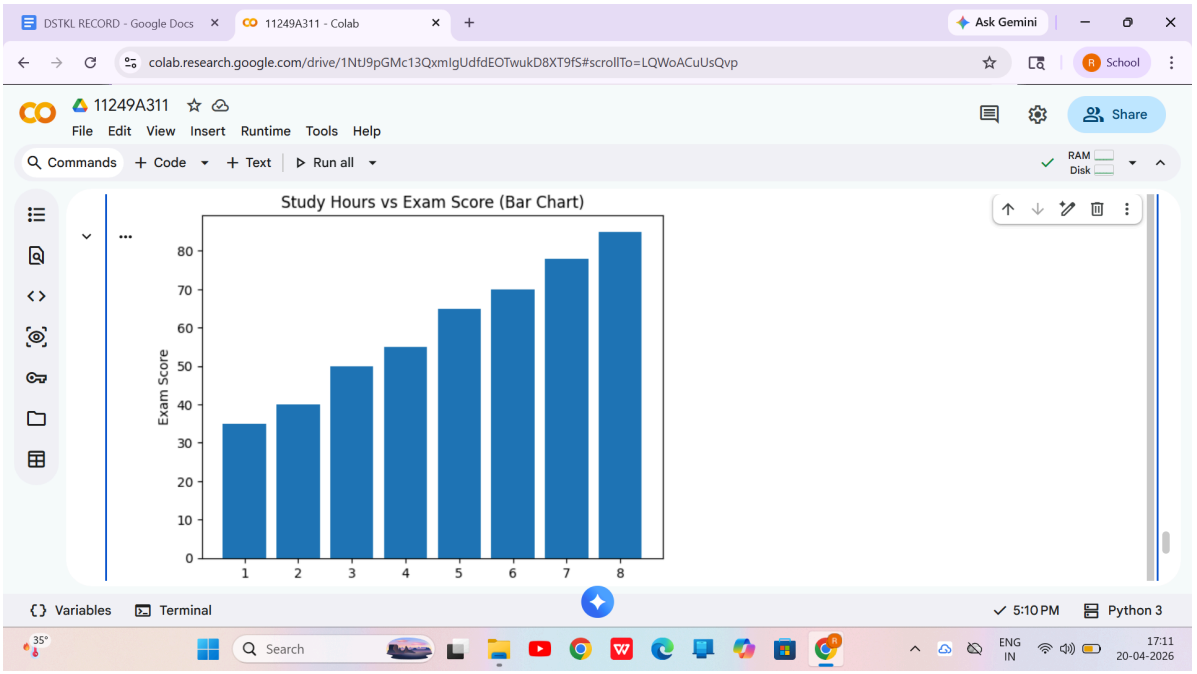
plt.show()

OUTPUT:






Exam Score Distribution (Histogram)
Number of Students
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
40
50
60
70
80


Study Hours vs Exam Score (Bar Chart)
Exam Score
80
70
60
50
40
30
20
10
0
1
2
3
4
5
6
7
8

SCIKIT LEARN(MACHINE LEARNING)

Aim:

To study and implement fundamental machine learning algorithms using Scikit-learn for building, training, and evaluating predictive models on various datasets.

STEP 1: IMPORT LIBRARIES

```
import pandas as pd

import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split

from sklearn.metrics import mean_squared_error
```

STEP 2: CREATE DATASET

```
data = {

"Student": ["S1", "S2", "S3", "S4", "S5",

"S6", "S7", "S8", "S9", "S10"],

"Classes_Attended": [30, 35, 40, 45, 50,

55, 60, 65, 70, 75],

"Internal_Marks": [35, 49, 38, 46, 50,

55, 60, 65, 70, 75]

}
```

```
df = pd.DataFrame(data)

print(df)
```

STEP 3: FEATURES + TARGET

```
X = df[["Classes_Attended"]] # Feature

y = df["Internal_Marks"] # Target
```

STEP 4: TRAIN - TEST SPLIT

```
X_train, X_test, y_train, y_test = train_test_split(

X, y, test_size=0.2, random_state=7

)
```

STEP 5: TRAIN MODEL

```
model = LinearRegression()

model.fit(X_train, y_train)

print(model)
```

STEP 6: PREDICTION

```
predictions = model.predict(X_test)
```

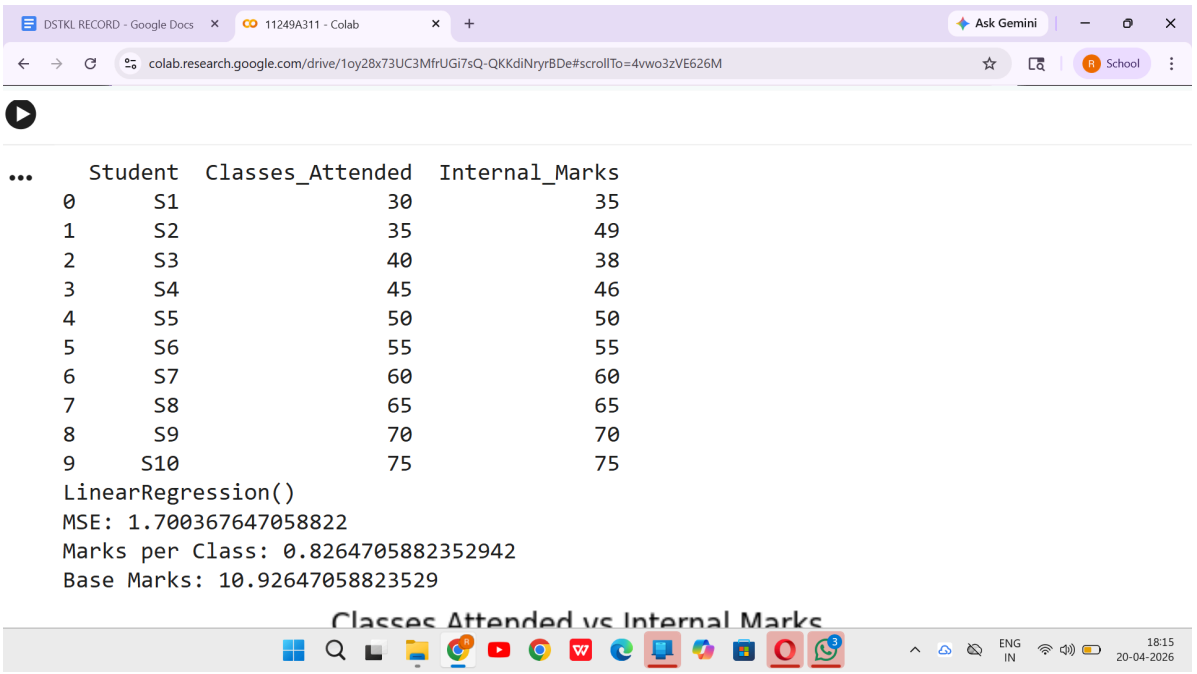
STEP 7: EVALUATION

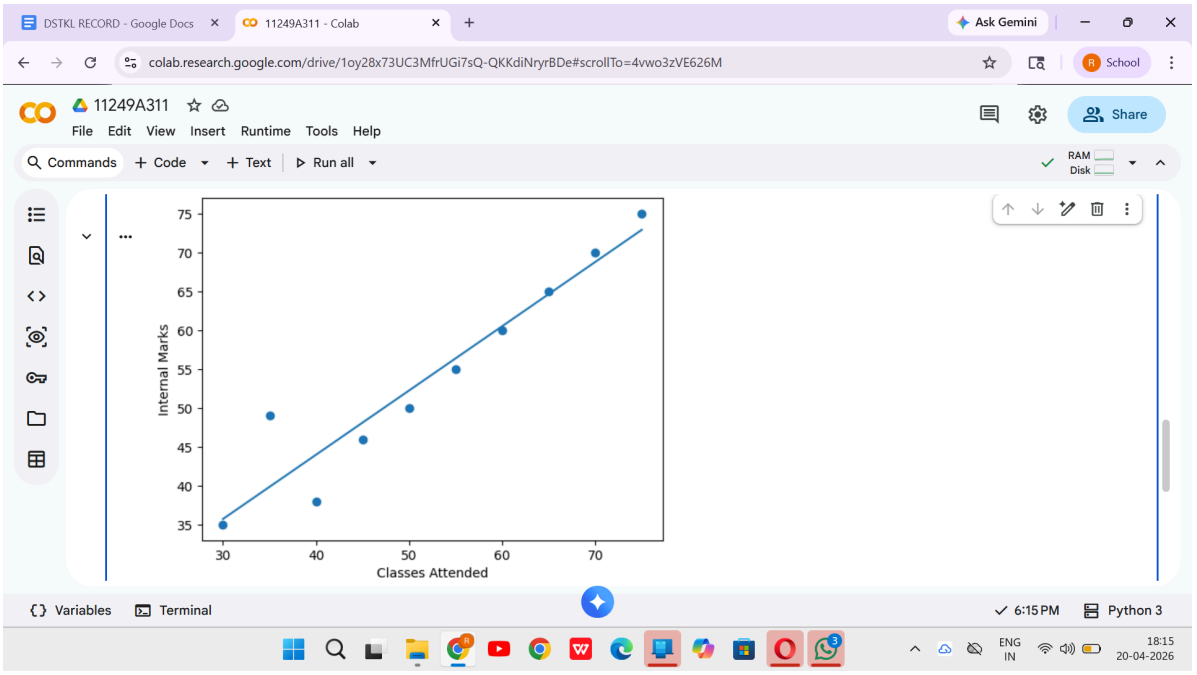
```
mse = mean_squared_error(y_test, predictions)

print("MSE:", mse)

print("Marks per Class:", model.coef_[0])

print("Base Marks:", model.intercept_)
```

STEP 8: VISUALIZATION

```
plt.scatter(X, y)

plt.plot(X, model.predict(X))

plt.xlabel("Classes Attended")

plt.ylabel("Internal Marks")

plt.title("Classes Attended vs Internal Marks")

plt.show()
```

OUTPUT:

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Ask Gemini
colab.research.google.com/drive/1oy28x73UC3MfrUGi7sQ-QKKdiNryrBDe#scrollTo=4vwo3zVE626M
School
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20-04-2026

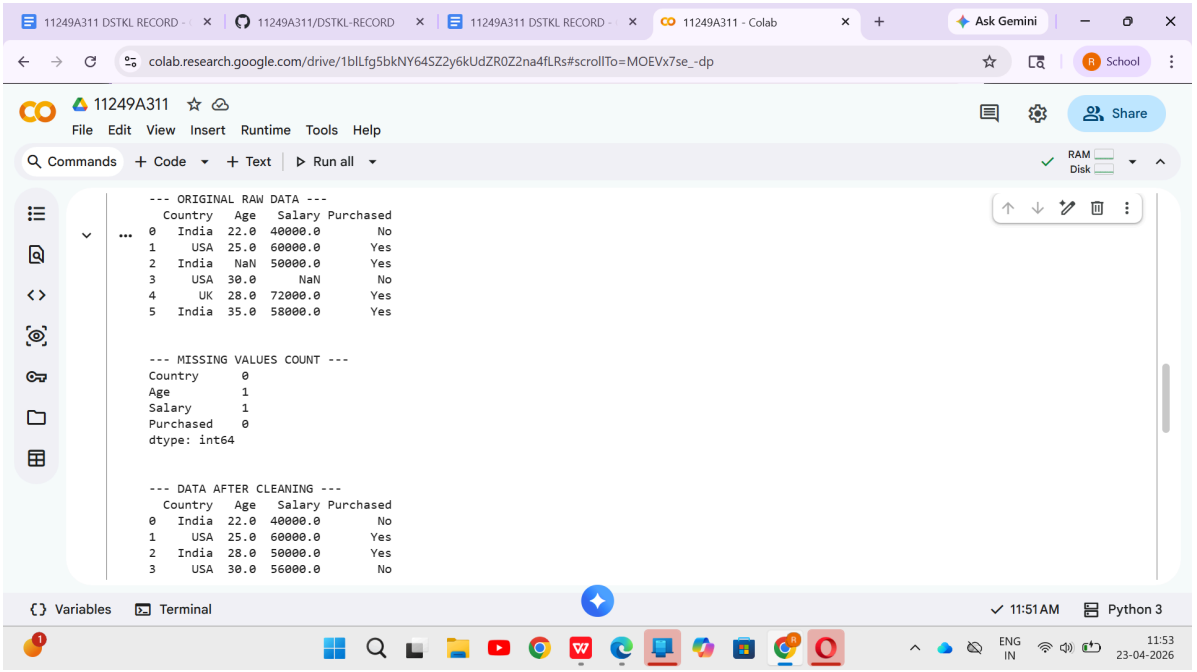
EX:-02

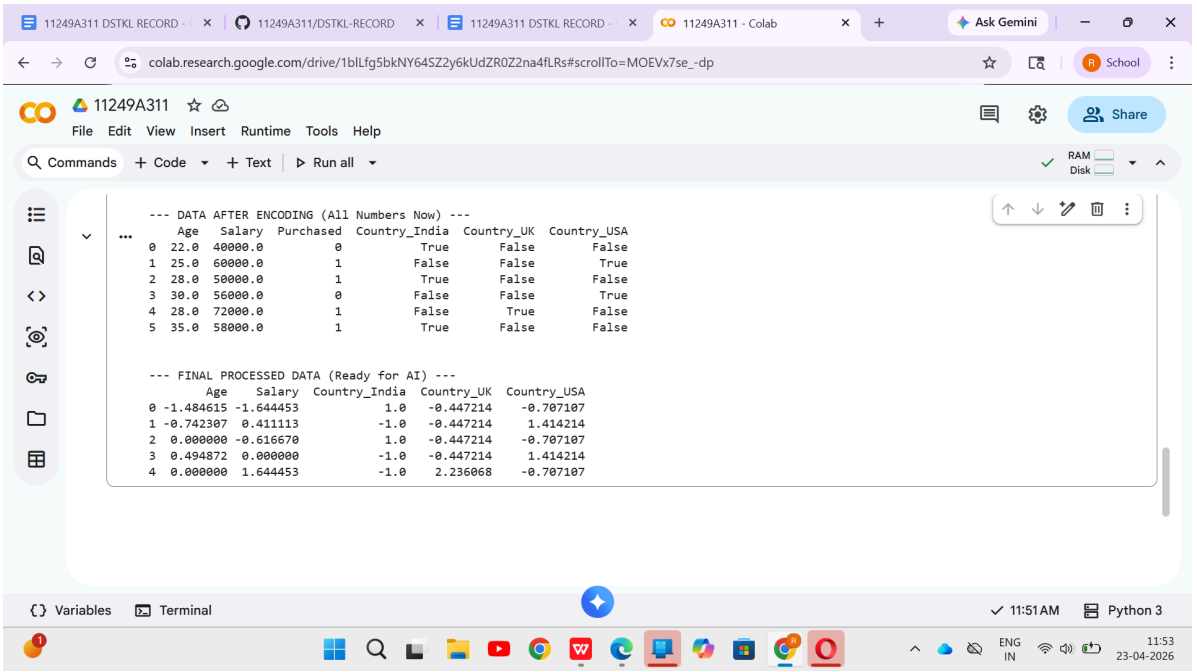
PERFORM DATA EXPLORATION and PRE-PROCCESING IN PYTHON

```
import pandas as pd

import numpy as np

from sklearn.preprocessing import StandardScaler

from sklearn.model_selection import train_test_split

# --- STEP 1: LOAD DATA (Creating a dummy dataset for the lab) ---

data = {

'Country': ['India', 'USA', 'India', 'USA', 'UK', 'India'],

'Age': [22, 25, np.nan, 30, 28, 35], # Note the 'nan' (missing value)

'Salary': [40000, 60000, 50000, np.nan, 72000, 58000],

'Purchased': ['No', 'Yes', 'Yes', 'No', 'Yes', 'Yes']

}

df = pd.DataFrame(data)

print("--- ORIGINAL RAW DATA ---")

print(df)

print("\n")

# --- STEP 2: INSPECT DATA ---
```

```
# Check for missing values

print("--- MISSING VALUES COUNT ---")

print(df.isnull().sum())

print("\n")

# --- STEP 3: CLEAN DATA (Handling Missing Values) ---

# Logic: Fill missing Age/Salary with the Average (Mean) of that column

df['Age'] = df['Age'].fillna(df['Age'].mean())

df['Salary'] = df['Salary'].fillna(df['Salary'].mean())

print("--- DATA AFTER CLEANING ---")

print(df)

print("\n")

# --- STEP 4: CONVERT TEXT TO NUMBERS (Encoding) ---

# Logic: Machines can't read 'India' or 'USA'. We convert them to 0s and 1s.

# We use 'One Hot Encoding' (get_dummies)

df_encoded = pd.get_dummies(df, columns=['Country'])

# For the Target column (Purchased), let's map Yes/No manually

df_encoded['Purchased'] = df_encoded['Purchased'].map({'Yes': 1, 'No': 0})

print("--- DATA AFTER ENCODING (All Numbers Now) ---")

print(df_encoded)print("\n")

# --- STEP 5: SCALE FEATURES ---
```

```
# Logic: Age is 20-30, Salary is 40000-60000. Salary dominates Age because it's bigger.

# We shrink them to the same scale.

# Separate Features (X) and Target (y)

X = df_encoded.drop('Purchased', axis=1)

y = df_encoded['Purchased']

# Scale

scaler = StandardScaler()

X_scaled = scaler.fit_transform(X)

print("--- FINAL PROCESSED DATA (Ready for AI) ---")

print(pd.DataFrame(X_scaled, columns=X.columns).head())
```

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--- DATA AFTER ENCODING (All Numbers Now) ---
Age Salary Purchased Country_India Country_UK Country_USA
0 22.0 40000.0 0 True False False
1 25.0 60000.0 1 False False True
2 28.0 50000.0 1 True False False
3 30.0 56000.0 0 False False True
4 28.0 72000.0 1 False True False
5 35.0 58000.0 1 True False False
--- FINAL PROCESSED DATA (Ready for AI) ---
Age Salary Country_India Country_UK Country_USA
0 -1.484615 -1.644453 1.0 -0.447214 -0.707107
1 -0.742307 0.411113 -1.0 -0.447214 1.414214
2 0.000000 -0.616670 1.0 -0.447214 -0.707107
3 0.494872 0.000000 -1.0 -0.447214 1.414214
4 0.000000 1.644453 -1.0 2.236068 -0.707107
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23-04-2026

3.IMPLEMENT REGULARISED LINEAR REGRESSION

Aim:

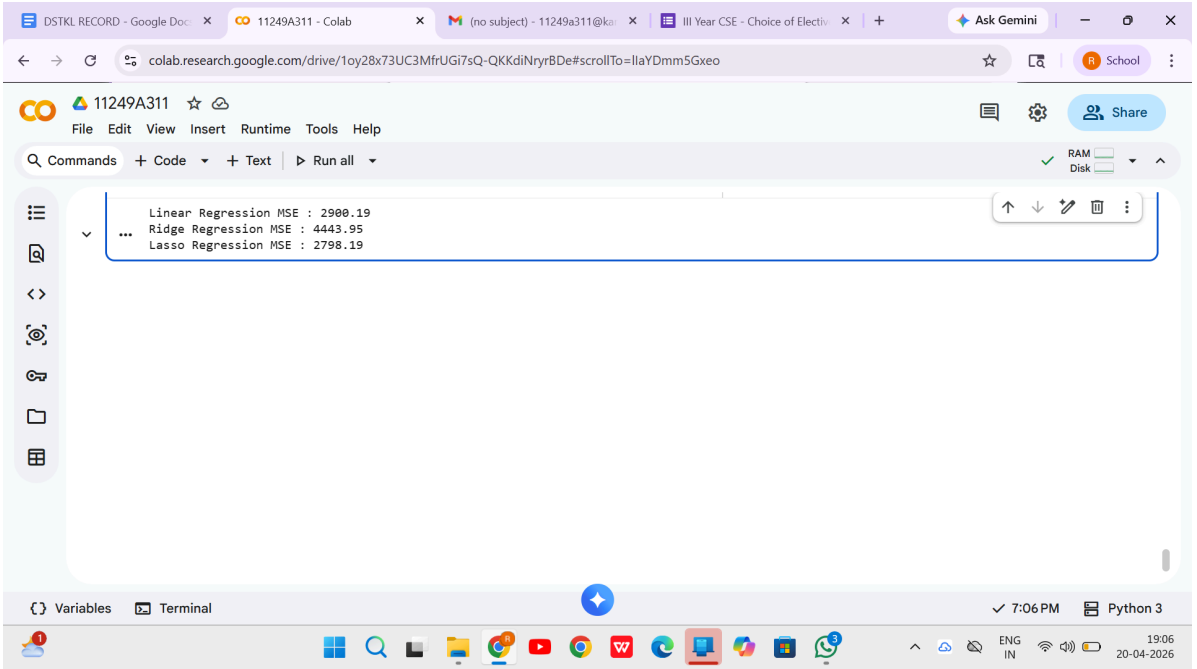
To implement **regularized linear regression** using Scikit-learn in order to build a regression model while reducing overfitting through regularization techniques such as Ridge (L2) and Lasso (L1), and to evaluate its performance on a given dataset.

PROGRAM:

```
from sklearn.linear_model import LinearRegression, Ridge, Lasso

from sklearn.metrics import mean_squared_error

from sklearn.datasets import load_diabetes

from sklearn.model_selection import train_test_split

# Load dataset

diabetes = load_diabetes()

# Split data

x_train, x_test, y_train, y_test = train_test_split(

    diabetes.data,

    diabetes.target,

    test_size=0.2,

    random_state=42

)

# Models
```

```
lr = LinearRegression().fit(x_train, y_train)

ridge = Ridge(alpha=10).fit(x_train, y_train)



lasso = Lasso(alpha=0.1).fit(x_train, y_train)

# Predictions and MSE

print(f"Linear Regression MSE : {mean_squared_error(y_test,
lr.predict(x_test)):.2f}")

print(f"Ridge Regression
MSE:{mean_squared_error(y_test,ridge.predict(x_test)):.2f}")

print(f"Lasso Regression MSE : {mean_squared_error(y_test,
lasso.predict(x_test)):.2f}")
```

OUTPUT:



4.IMPLEMENT NAIVE BAYES CLASSIFIER

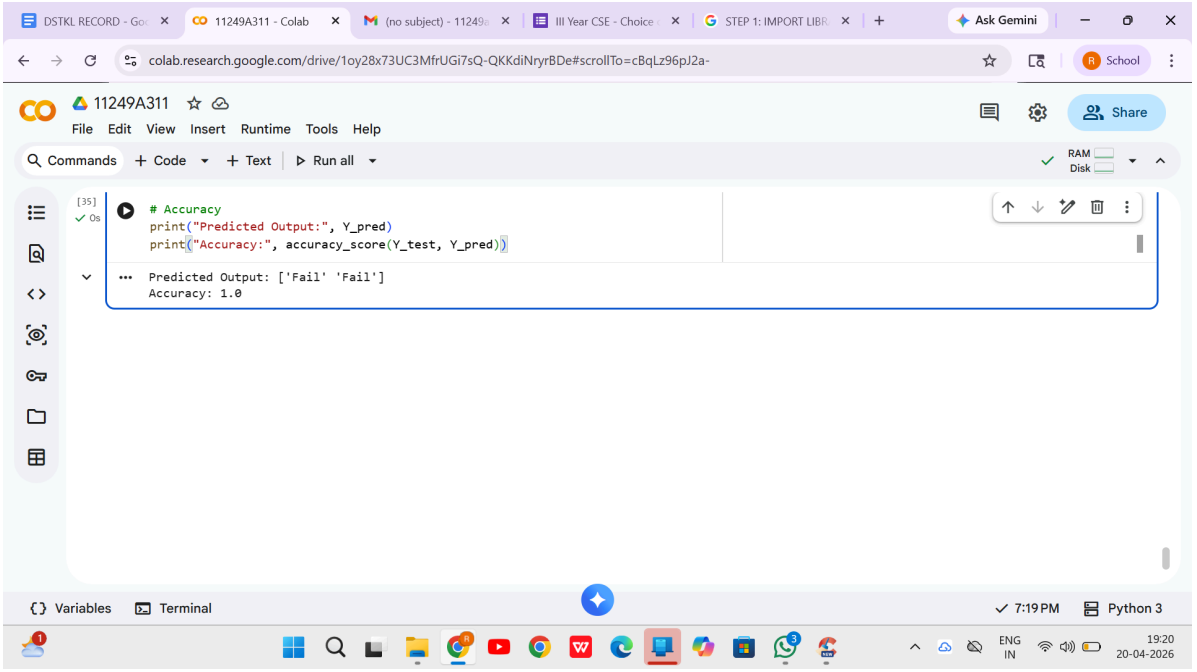
Aim:

To implement the **Naive Bayes classifier** using Scikit-learn for building a probabilistic classification model based on Bayes' theorem, and to evaluate its performance on a given dataset.

```
import pandas as pd

from sklearn.model_selection import train_test_split

from sklearn.naive_bayes import GaussianNB

from sklearn.metrics import accuracy_score

# Dataset

data = {

    'StudyHours': [5, 2, 8, 1, 6, 3, 7, 4],

    'Attendance': [80, 50, 90, 40, 85, 60, 88, 65],

    'Result': ['Pass', 'Fail', 'Pass', 'Fail', 'Pass', 'Fail', 'Pass', 'Fail']

}

# Create DataFrame

df = pd.DataFrame(data)

# Features and Target

X = df[['StudyHours', 'Attendance']]
```

```
Y = df['Result']

# Split dataset

X_train, X_test, Y_train, Y_test = train_test_split(

    X, Y, test_size=0.25, random_state=42

)

# Model

model = GaussianNB()

# Train model

model.fit(X_train, Y_train)

# Prediction

Y_pred = model.predict(X_test)

# Accuracy

print("Predicted Output:", Y_pred)

print("Accuracy:", accuracy_score(Y_test, Y_pred))
```

OUTPUT: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# Accuracy
print("Predicted Output:", Y_pred)
print("Accuracy:", accuracy_score(Y_test, Y_pred))
Predicted Output: ['Fail' 'Fail']
Accuracy: 1.0
Variables
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20-04-2026

5.IMPLEMENT REGULARIZED LOGISTIC REGRESSION

Aim:

To implement **regularized logistic regression** using Scikit-learn in order to build a classification model while controlling overfitting through regularization techniques such as L1 and L2 penalties, and to evaluate its performance on a given dataset.

 # 1. Import libraries

```
import numpy as np

import matplotlib.pyplot as plt

from sklearn.datasets import load_breast_cancer

from sklearn.model_selection import train_test_split

from sklearn.preprocessing import StandardScaler

from sklearn.linear_model import LogisticRegression

from sklearn.metrics import accuracy_score
```

 # 2. Load Dataset

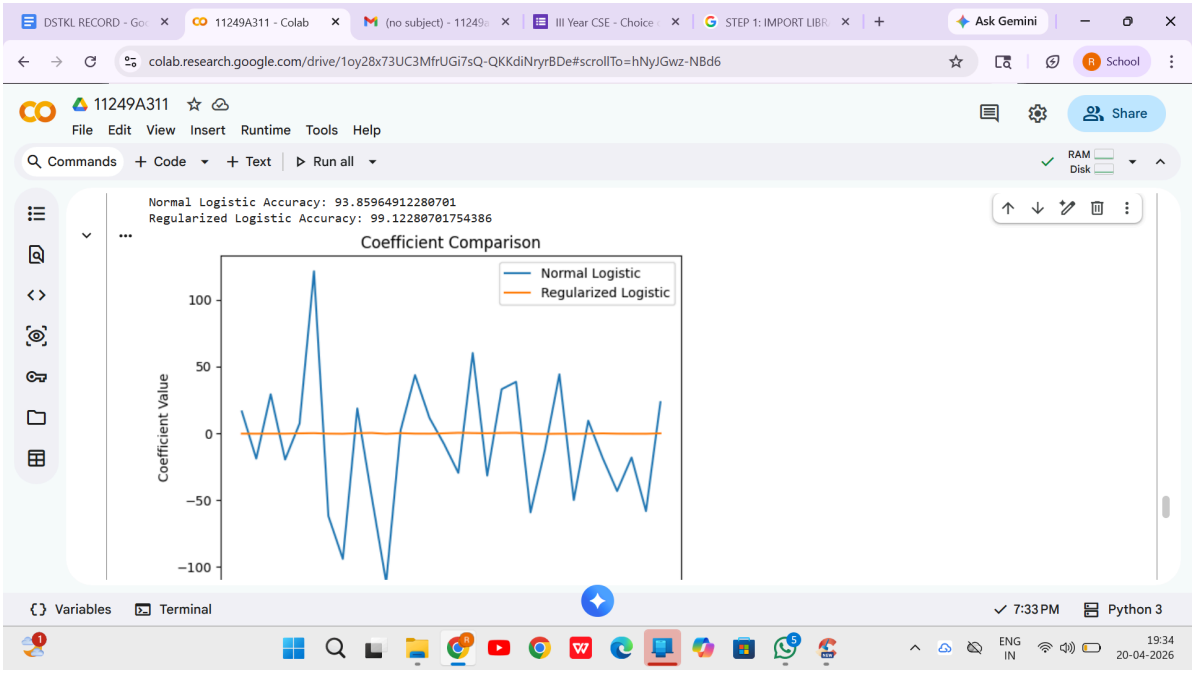
```
data = load_breast_cancer()

X = data.data

y = data.target
```

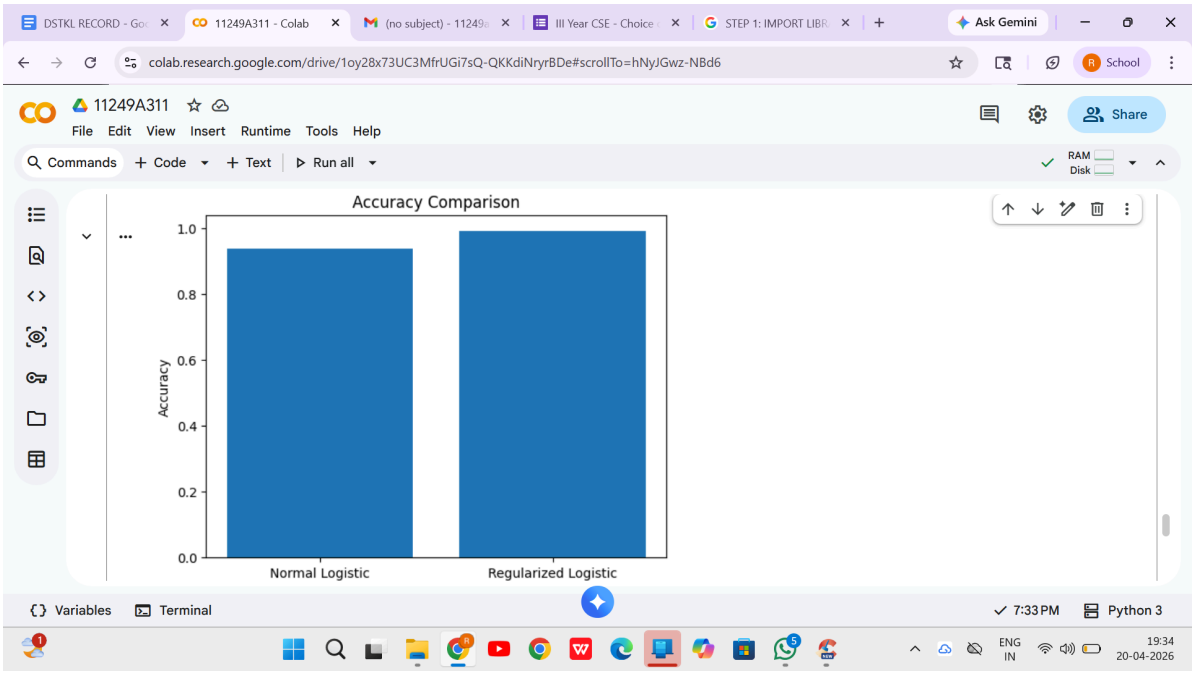
 # 3. Split data

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
# 4. Scaling
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
# 5. Normal Logistic Regression (very large C)
normal_model = LogisticRegression(C=1e6, penalty='l2', solver='liblinear')
normal_model.fit(X_train_scaled, y_train)
normal_pred = normal_model.predict(X_test_scaled)
normal_acc = accuracy_score(y_test, normal_pred)
# 6. Regularized Logistic Regression
regularized_model = LogisticRegression(C=0.1, penalty='l2',
solver='liblinear')
regularized_model.fit(X_train_scaled, y_train)
reg_pred = regularized_model.predict(X_test_scaled)
reg_acc = accuracy_score(y_test, reg_pred)
# 7. Print Results
```

```
print("Normal Logistic Accuracy:", normal_acc * 100)

print("Regularized Logistic Accuracy:", reg_acc * 100)

# 8. Compare Coefficients

plt.figure()

plt.plot(normal_model.coef_.flatten(), label='Normal Logistic')

plt.plot(regularized_model.coef_.flatten(), label='Regularized Logistic')

plt.legend()

plt.title("Coefficient Comparison")

plt.xlabel("Feature Index")

plt.ylabel("Coefficient Value")

plt.show()

# 9. Accuracy Comparison Bar Graph

plt.figure()

models = ['Normal Logistic', 'Regularized Logistic']

accuracies = [normal_acc, reg_acc]

plt.bar(models, accuracies)

plt.title("Accuracy Comparison")

plt.ylabel("Accuracy")

plt.show()
```

OUTPUT:


Normal Logistic Accuracy: 93.85964912280701
Regularized Logistic Accuracy: 99.12280701754386
Coefficient Comparison
Normal Logistic
Regularized Logistic
Coefficient Value
100
50
0
−50
−100


Accuracy Comparison
Accuracy
1.0
0.8
0.6
0.4
0.2
0.0
Normal Logistic
Regularized Logistic

6. Build models using different ensembling techniques

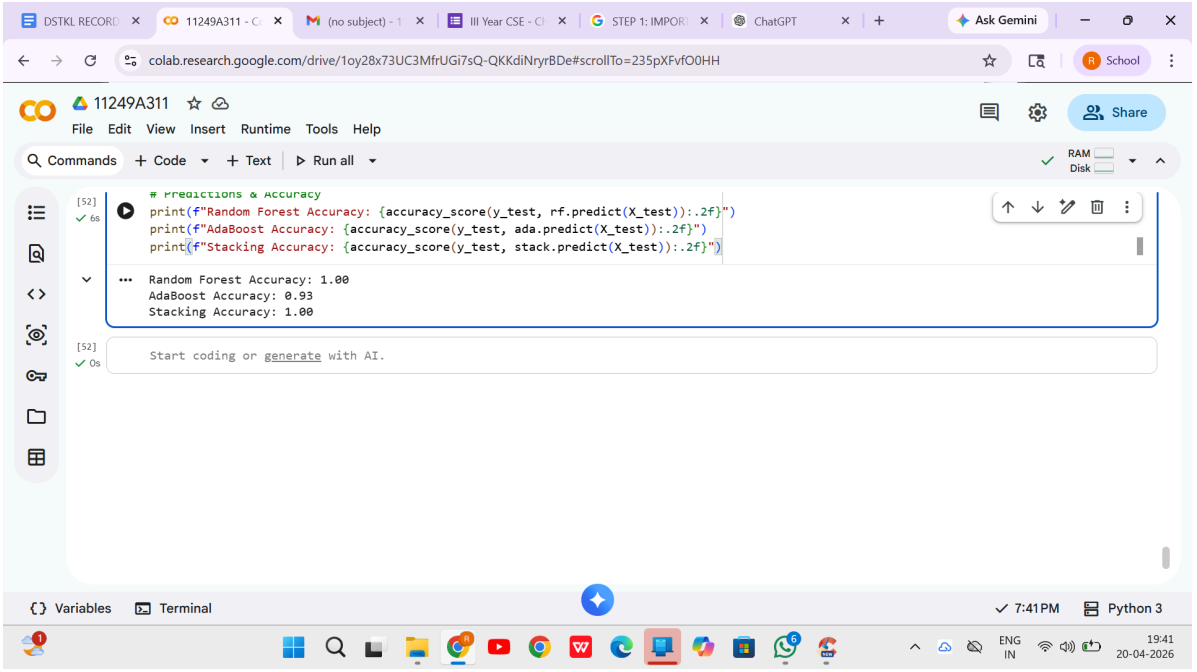
Aim:

To implement and compare different ensemble learning techniques using Scikit-learn—specifically Random Forest, AdaBoost, and Stacking classifiers—for building predictive models and evaluating their performance on a dataset.

```
from sklearn.ensemble import RandomForestClassifier, AdaBoostClassifier,
StackingClassifier

from sklearn.svm import SVC

from sklearn.linear_model import LogisticRegression

from sklearn import datasets

from sklearn.model_selection import train_test_split

from sklearn.metrics import accuracy_score

# Load dataset

iris = datasets.load_iris()

# Split data

X_train, X_test, y_train, y_test = train_test_split(

    iris.data, iris.target, test_size=0.2, random_state=42

)

# Random Forest

rf = RandomForestClassifier(n_estimators=100)

rf.fit(X_train, y_train)
```

```
# AdaBoost
ada = AdaBoostClassifier(n_estimators=100)
ada.fit(X_train, y_train)
# Stacking
estimators = [
    ('rf', RandomForestClassifier(n_estimators=100)),
    ('svc', SVC(probability=True))
]
stack = StackingClassifier(
    estimators=estimators,
    final_estimator=LogisticRegression()
)
stack.fit(X_train, y_train)
# Predictions & Accuracy
print(f"Random Forest Accuracy: {accuracy_score(y_test, rf.predict(X_test)):.2f}")
print(f"AdaBoost Accuracy: {accuracy_score(y_test, ada.predict(X_test)):.2f}")
print(f"Stacking Accuracy: {accuracy_score(y_test, stack.predict(X_test)):.2f}")
```

OUTPUT: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# Predictions & Accuracy
print(f"Random Forest Accuracy: {accuracy_score(y_test, rf.predict(X_test)):.2f}")
print(f"AdaBoost Accuracy: {accuracy_score(y_test, ada.predict(X_test)):.2f}")
print(f"Stacking Accuracy: {accuracy_score(y_test, stack.predict(X_test)):.2f}")
Random Forest Accuracy: 1.00
AdaBoost Accuracy: 0.93
Stacking Accuracy: 1.00
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7.BUILDING MODELS USING DECISION TREES

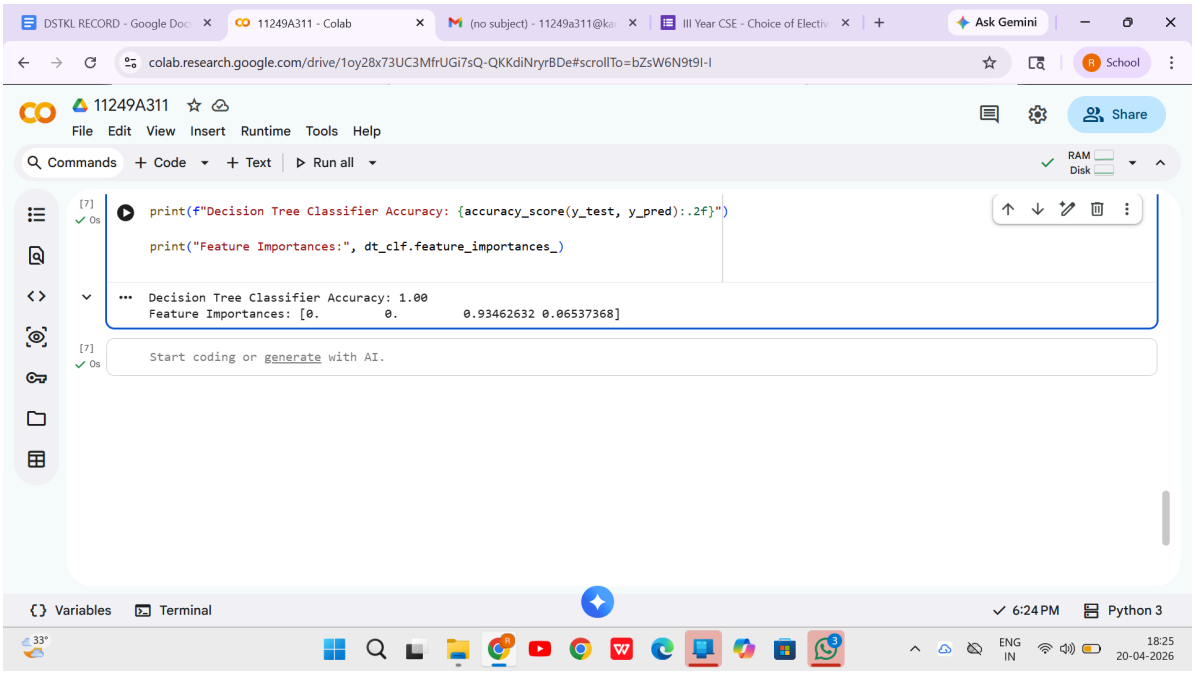
Objective:

To implement the Decision Tree algorithm for classification and regression.from sklearn.tree import DecisionTreeClassifier

PROGRAM:

```
from sklearn import datasets

from sklearn.model_selection import train_test_split

from sklearn.tree import DecisionTreeClassifier

from sklearn.metrics import accuracy_score

iris = datasets.load_iris()

X_train, X_test, y_train, y_test = train_test_split(

    iris.data, iris.target, test_size=0.2, random_state=42

)

dt_clf = DecisionTreeClassifier(max_depth=3)

dt_clf.fit(X_train, y_train)

y_pred = dt_clf.predict(X_test)

print(f"Decision Tree Classifier Accuracy: {accuracy_score(y_test, y_pred):.2f}")

print("Feature Importances:", dt_clf.feature_importances_)
```

OUTPUT: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RAM
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print(f"Decision Tree Classifier Accuracy: {accuracy_score(y_test, y_pred):.2f}")
print("Feature Importances:", dt_clf.feature_importances_)
Decision Tree Classifier Accuracy: 1.00
Feature Importances: [0. 0. 0.93462632 0.06537368]
Start coding or generate with AI.
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20-04-2026

8) Support Vector Machine (SVM) with Different Kernels

Support Vector Machine is a supervised machine learning algorithm used for classification and regression problems.

SVM finds the best hyperplane that separates data points of different classes, Example:

Class A • Class B

SVM finds a line (in 2D) or hyperplane (in higher dimensions) that separates these classes with maximum margin.

Key terms:

Hyperplane

Decision boundary separating classes.

Support Vectors

The data points closest to the decision boundary.

Margin

Distance between hyperplane and nearest data points.

Kernel Functions

separate classes,

Sometimes data is not linearly separable. Kernels transform data into a higher dimension to

Common kernels:

1. Linear Kernel

Used v/hen data is linearly separable,

2 Polynomial Kernel

Creates curved decision boundaries,

3 RBF Kernel (Radial Basis Function)

Most commonly used kernel.

Algorithm Steps

1 Load dataset

2 Split dataset into training and testing

3 Train SVM model with different kernels

4 Predict output

5 Compare accuracy

Coding:

```
# Import libraries

import pandas as pa

from sklearn. model selection import train test split from sklearn.svm import SVC

from sklearn metrics import accuracy score from sklearn.datasets import load iris

# Load dataset

data = load iris()

X = data.data

y = data target

# Split dataset

X train, X test, y_train, y test = train test split(X, y, test size=0.2)

# Linear Kernel

mode inear = SVC(kernel-'linear')

model linear fit(X train, y_train)

pred linear = model linear predict(X. test)

acc linear - accuracy score(y test pred linear)

print( Linear Kernel Accuracy:", acc linear)

# Polynomial Kernel

model poly - SVC(kernel='poly', degree-3)

model poly fit(X train, y_train)

pred poly - model poly predict(X test) acc poly - accuracy score(y_test, pred poly)

print("Polynomial Kernel Accuracy:", acc poly)
```

RBF Kernel

model rbf - SVC(kernel=rbf)

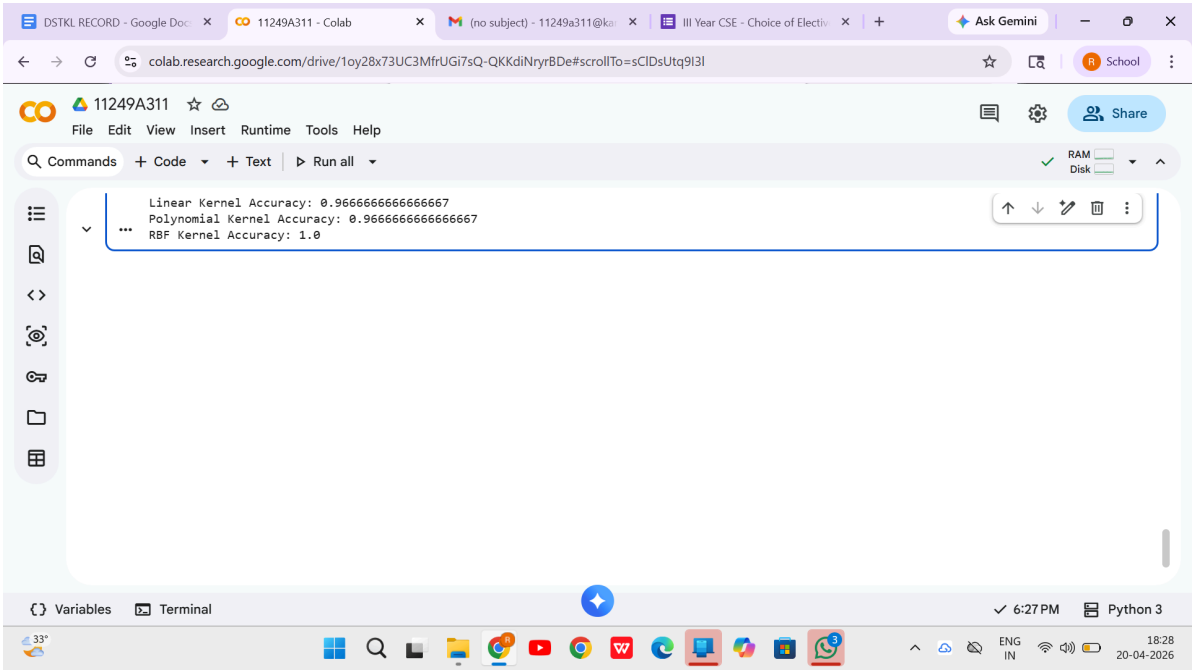
model rbffit(X train, y_ train)

pred rbf = model rbf predict(X. test)

acc rbf - accuracy scorely test, pred rbf)

print("RBF Kernel Accuracy:*, acc rbf)

Output:



9) Implement K-Nearest Neighbors (KNN) Classification

To implement the K-Nearest Neighbors (KNN) algorithm to classify mobile phones into price categories using Python.

Dataset:

Mobile Price Classification Dataset. It classifies mobile phones into 4 price ranges based on features like RAM, battery power, etc.

Kaggle Dataset:

This dataset is good because:

• Only numerical features (easy for KNN)

• Multi-class classification

• Clean CSV dataset suitable for engineering lab experiments

Algorithm

1. Import required libraries.

2. Load the dataset.

3. Split dataset into features (X) and target (y).

4. Divide data into training and testing sets.

5. Normalize features using StandardScaler.

6. Train the KNN classifier.

7. Predict results for test data.

8. Evaluate accuracy.

Coding:

```
# Step 1: Import libraries
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
# Step 2: Load dataset
data = pd.read_csv("mobile_price.csv")
# Step 3: Define features and target
X = data.drop("price_range", axis=1)
y = data["price_range"]
# Step 4: Train-test split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
# Step 5: Feature scaling
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

 # Step 6: Create KNN model

knn = KNeighborsClassifier(n_neighbors=5)

 # Step 7: Train model

knn.fit(X_train, y_train)

 # Step 8: Prediction

y_pred = knn.predict(X_test)

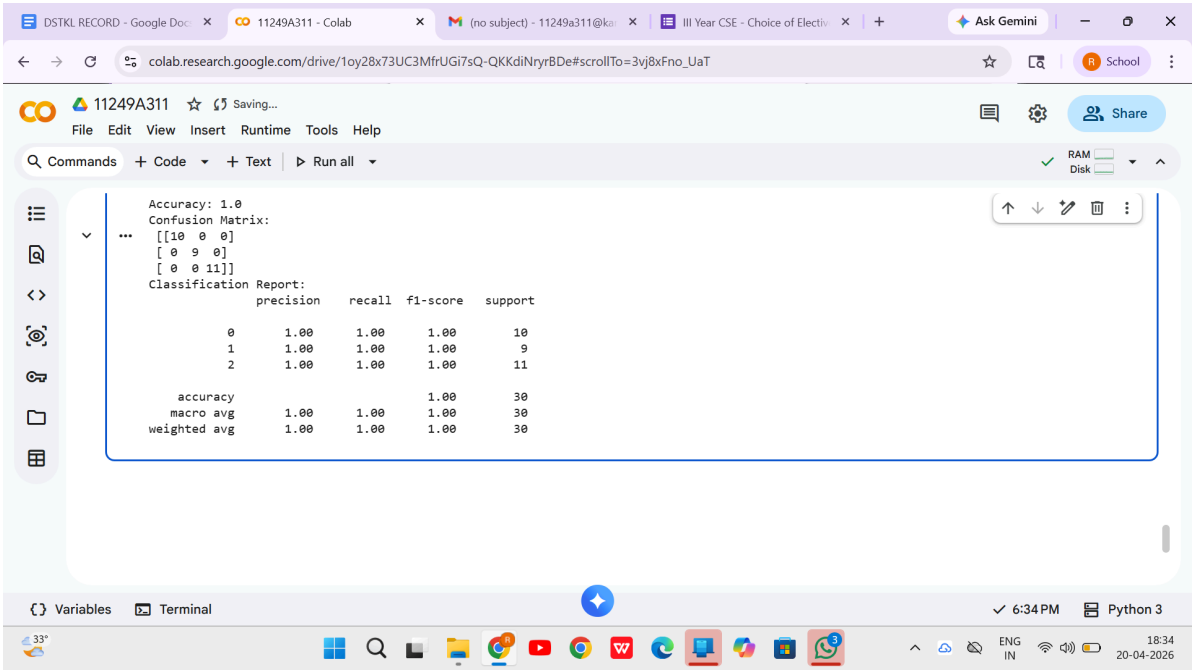
 # Step 9: Evaluation

print("Accuracy:", accuracy_score(y_test, y_pred))

print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred))

print("Classification Report:\n", classification_report(y_test, y_pred))

OUTPUT:



10.) K-Means Clustering with PCA and Elbow Method

To perform clustering using K-Means algorithm after applying PCA (Principal Component Analysis) and determine the optimal K value using the Elbow Method.

Kaggle Dataset

Mall Customer Segmentation Dataset

• Dataset: Mall Customers

• Records: 200 customers

• Features:

• CustomerID

o Gender

• Age

• Annual Income (k$)

• Spending Score (1-100)

This dataset is widely used for customer segmentation using K-Means clustering and is ideal for student labs.

Typical clustering:

• High income - high spending

• High income - low spending

• Low income - high spending

• Low income - low spending

PROGRAM :

```
# Step 1: Import libraries

import pandas as pd

import matplotlib.pyplot as plt

from sklearn.preprocessing import StandardScaler

from sklearn.decomposition import PCA

from sklearn.cluster import KMeans

# Step 2: Load dataset

data = pd.read_csv("Mall_Customers.csv")

# Step 3: Select numerical features

X = data[['Age', 'Annual Income (k$)', 'Spending Score (1-100)']]

# Step 4: Feature Scaling

scaler = StandardScaler()

X_scaled = scaler.fit_transform(X)

# Step 5: Apply PCA

pca = PCA(n_components=2)

X_pca = pca.fit_transform(X_scaled)

print("Explained Variance Ratio:", pca.explained_variance_ratio_)

# Step 6: Elbow Method

wcss = [ ]
```

```
for i in range(1, 11):
    kmeans = KMeans(n_clusters=i, random_state=0)
    kmeans.fit(X_pca)
    wcss.append(kmeans.inertia_)
```

 # Step 7: Plot Elbow Graph

```
plt.plot(range(1, 11), wcss, marker='o')
plt.title("Elbow Method")
plt.xlabel("Number of Clusters (K)")
plt.ylabel("WCSS")
plt.show()
```

 # Step 8: Apply KMeans with optimal K (e.g., K=5)

```
kmeans = KMeans(n_clusters=5, random_state=0)
clusters = kmeans.fit_predict(X_pca)
```

 # Step 9: Plot Clusters

```
plt.scatter(X_pca[:, 0], X_pca[:, 1], c=clusters, cmap='viridis')
plt.title("K-Means Clustering with PCA")
plt.xlabel("PCA Component 1")
plt.ylabel("PCA Component 2")
plt.show()
```

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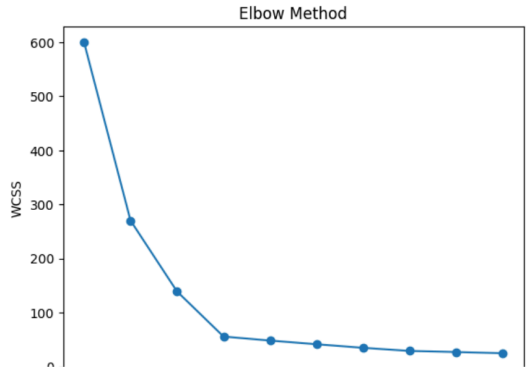
```
Explained Variance Ratio: [0.76969335 0.23030665]
```



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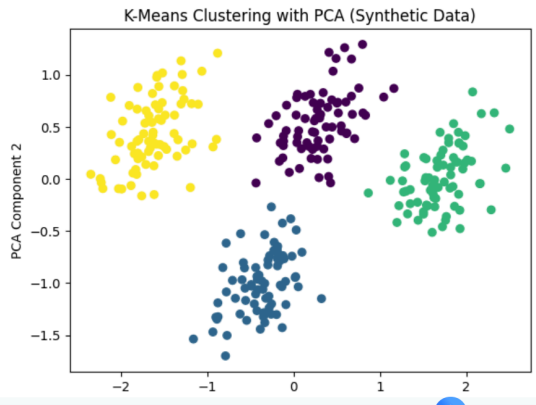

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